clk100mhz

st_wrt

data_in[11..0]

Add1 +

CIN 1'h0

A[5..0] B[5..0] 6'h1

OUT[5..0]

count~[5..0]

0 6'h0 1

Selector3

SEL[3..0] DATA[3..0] OUT

Selector4

SEL[3..0] DATA[3..0] OUT

Selector2

SEL[3..0] DATA[3..0] OUT

Selector5

SEL[3..0] DATA[3..0] OUT

Selector6

SEL[3..0] DATA[3..0] OUT

Selector7

SEL[3..0] DATA[3..0] OUT

count[5..0] D
CLK
CLRN

SCLR

6'h0

Q

LessThan0 <

CIN 1'h0

A[5..0] B[5..0] 6'h20

OUT dac_init 0 1'h1 1

dac_init

D
CLK

CLRN

ENA
SCLR

1'h0

Q

state~7
1'h0 state
clk
count[5..0]
dac_init
reset

dac_data
idle_dac
init_dac
send_data

Add0 +

CIN 1'h0

A[5..0] B[5..0] 6'h1

OUT[5..0]

data~[31..0]

01

clkdiv~[5..0]

01 6'h0

data[31..0] D
CLK
ENA
SCLR

32'h0

Q

clkdiv[5..0] D
CLK
SCLR

6'h0

Q

Mux0

SEL[4..0] DATA[31..0] 32'h80000010 OUT
Mux1

SEL[4..0] DATA[31..0] OUT

Equal0 =

A[5..0] B[5..0] 6'h31 OUT

mosi~[1..0]

01

cs

done~1

clk1mhz D
CLK
ENA
SCLR

1'h0

Q

Selector1

SEL[2..0] DATA[2..0] OUT

Selector0

SEL[1..0] DATA[1..0] OUT

Selector8

SEL[2..0] DATA[2..0] OUT

mosi~reg0 D
CLK
CLRN

SCLR

1'h0

Q

cs~reg0

D
CLK

PRE

SCLR

1'h0

Q

done~reg0 D
CLK
CLRN

SCLR

1'h0

Q

done~0

0 1'h1 1

mosi

cs

sclk done

0:11

4->DATA[1]235 DATA[1]245<-3 DATA[1]225<-5 DATA[1]255<-2 DATA[1]265<-1 DATA[1]275<-0

4:0

4:0

1,1 2,2 0,0 3,3 4,4 5,5

0:1